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Saveetha Institute of Medical And Technical Sciences
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AI-ENABLED RFID SENSOR TECHNOLOGY FOR INTELLIGENT IDENTIFICATION AND TRACKING

MLA0104 - ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

C04 –AT 3 (TECHNICAL PRESENTATION)

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INTRODUCTION TO RFID TECHNOLOGY

- RFID stands for Radio Frequency Identification.
- It uses radio-frequency waves to identify and track objects.
- RFID enables contactless and automatic identification.
- It can read information without direct physical contact.
- Unlike barcodes, RFID generally does not require direct line-of-sight.
- RFID is widely used in IoT, automation and intelligent monitoring systems.

COMPONENTS OF RFID SYSTEM

An RFID system consists of the following key components working together to identify, capture, process and store data.

1. RFID TAG | **2. ANTENNA** | **3. RFID READER** | **4. CONTROLLER / COMPUTER** | **5. DATABASE** | **6. SOFTWARE / APPLICATION**

1. RFID TAG: Stores unique ID and other data.
2. ANTENNA: Transmits and receives radio-frequency signals between tag and reader.
3. RFID READER: Communicates with tags, reads their data and sends to the system.
4. CONTROLLER / COMPUTER: Processes the data, applies business logic and manages operations.
5. DATABASE: Stores tag information, transaction history and related data securely.
6. SOFTWARE / APPLICATION: Displays information, generates reports and provides user interface.

1. RFID TAG

- Contains a microchip and an antenna.
- Stores a unique identification (ID) and other data.
- Can be Passive, Active or Semi-passive.
- Attached to or embedded in objects.

2. ANTENNA

- Acts as an interface between tag and reader.
- Generates RF field to power the tag (in passive tags).
- Receives the response signal from the tag.

3. RFID READER

- Sends radio-frequency signals to the antenna.
- Powering the tag and communicates with it.
- Reads data from the tag through the antenna.
- Decodes the data and forwards it to the controller/computer.

4. CONTROLLER / COMPUTER

- Receives data from the reader.
- Filters duplicate reads and validates information.
- Executes business logic and sends data to database/software.

5. DATABASE

- Stores all RFID data in an organized manner.
- Maintains records of tags, time, location, and actions.
- Ensures data security and fast retrieval.

6. SOFTWARE / APPLICATION

- Provides user interface for monitoring and management.
- Generates reports, alerts and analytics.
- Integrates RFID data with other enterprise systems.

DATA FLOW: RFID TAG → ANTENNA → RFID READER → CONTROLLER / COMPUTER → DATABASE → SOFTWARE / APPLICATION

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WORKING PRINCIPLE OF RFID

- The RFID reader generates and transmits a radio-frequency signal.
- The RFID tag receives the signal through its antenna.
- A passive RFID tag obtains energy from the reader's electromagnetic field.
- The tag sends its stored identification data back to the reader.
- The reader receives and decodes the tag information.
- The computer processes the received data for identification, tracking or further analysis.

RFID: SIGNIFICANCE AND DEPLOYMENT AREAS

KEY ADVANTAGES:

- Fast and contactless identification.
- No direct line-of-sight is generally required.
- Supports automated and real-time tracking.
- Can support reading multiple tags.
- Reduces manual data-entry errors.

MAJOR DEPLOYMENT AREAS:

- Retail Stores – Inventory and product tracking.
- Hospitals – Patient and medical-equipment identification.
- Industries – Tracking tools, components and assets.
- Warehouses – Stock and logistics management.
- Security Systems – Access control and identification.

APPLICATIONS & AI INTEGRATION

MAJOR APPLICATIONS:

- **Smart Retail:** Inventory tracking, stock monitoring and automated checkout.
- **Healthcare:** Patient identification, medicine and medical-equipment tracking.
- **Logistics:** Real-time tracking of goods, shipments and warehouse inventory.
- **Access & Security:** Contactless entry, attendance and authorization systems.
- **Industry 4.0:** Asset, tool and production-process tracking.

AI/ML INTEGRATION:

- **Anomaly Detection:** Identifies unusual tag activity or unauthorized access.
- **Classification:** Recognizes and categorizes RFID events and movement patterns.
- **Predictive Analytics:** Forecasts inventory requirements and asset movement.
- **Intelligent Tracking:** Analyzes RFID data for smarter monitoring and localization.
- **Automated Decisions:** Enables real-time alerts and intelligent system responses.

Thank you